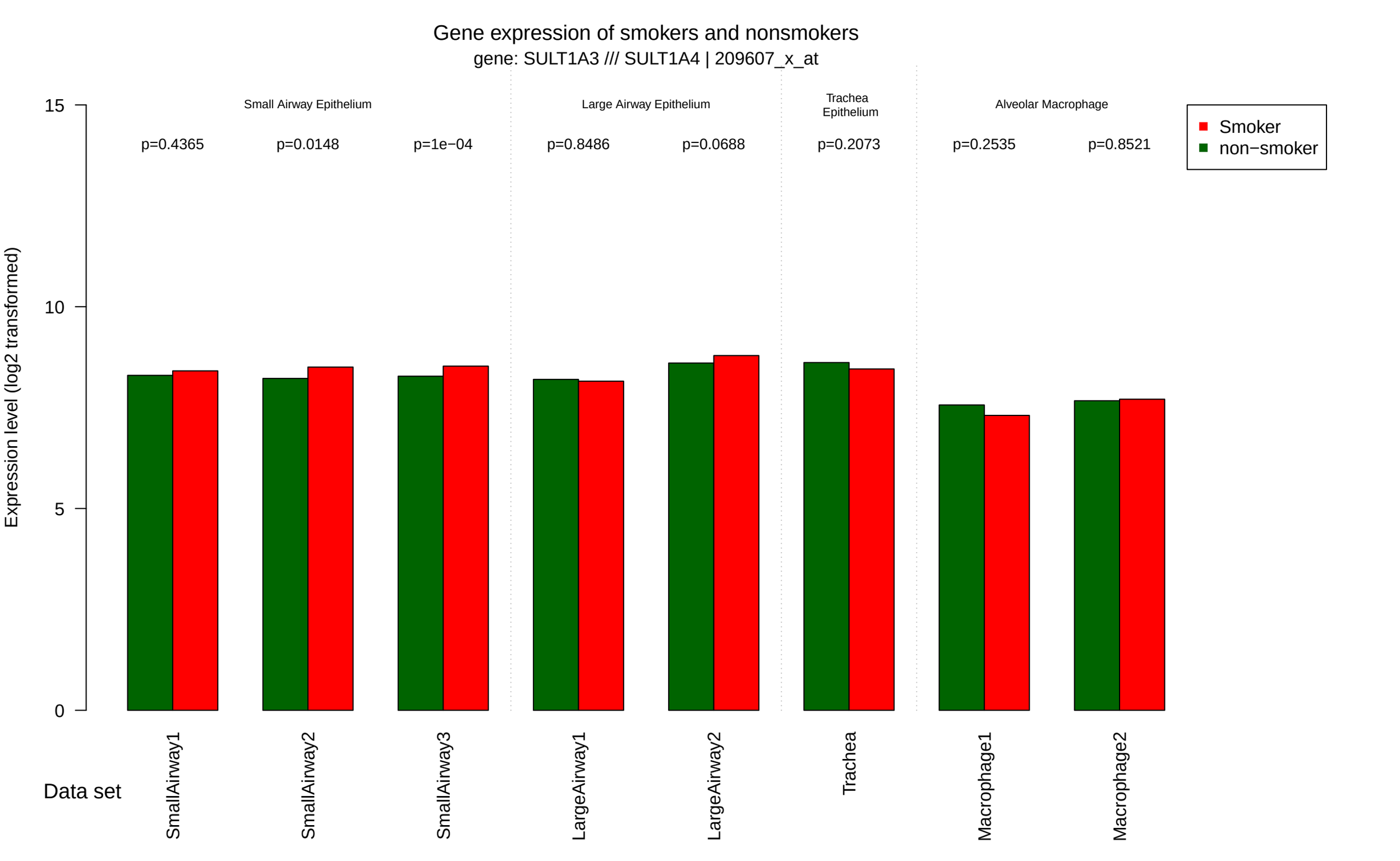
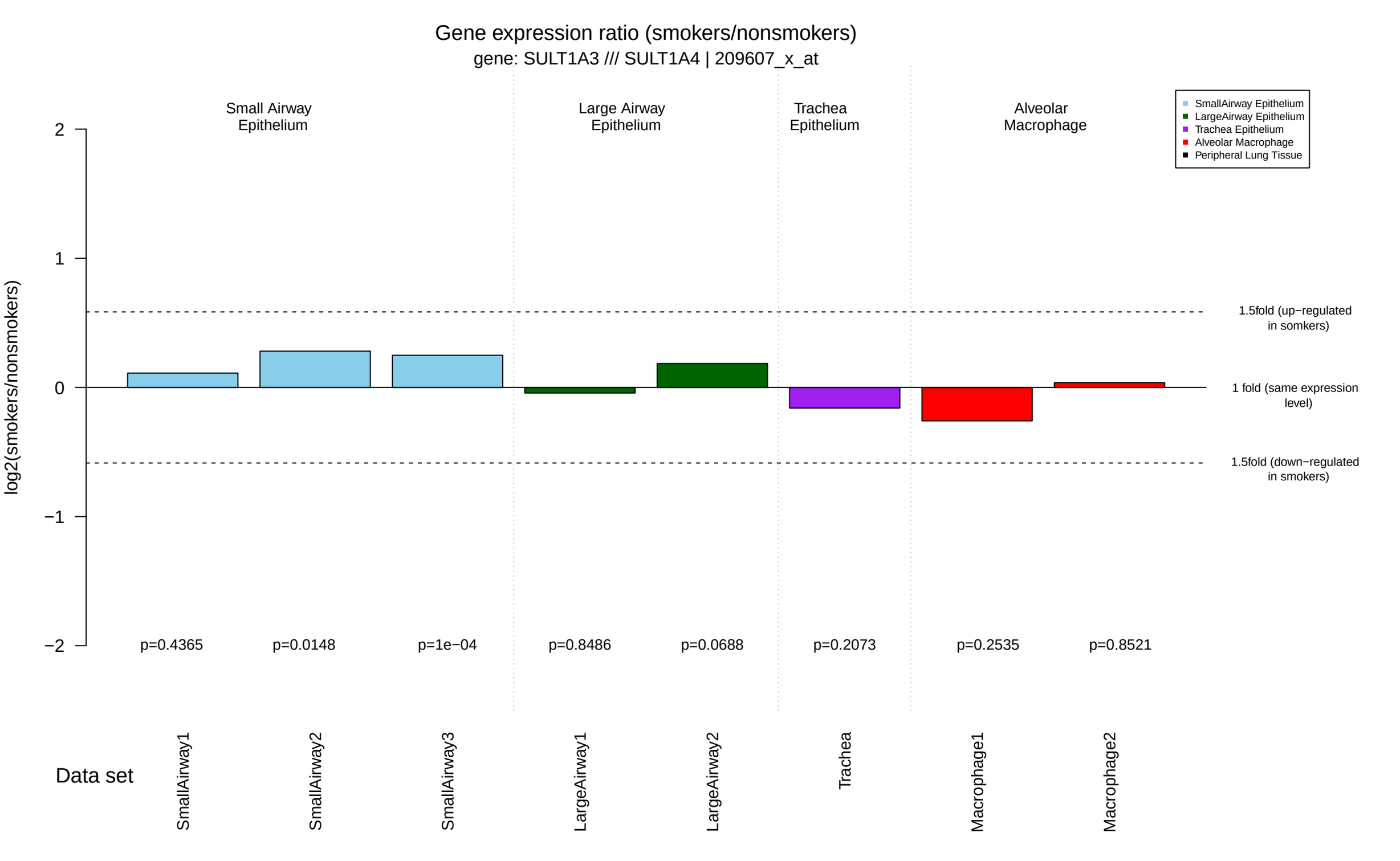
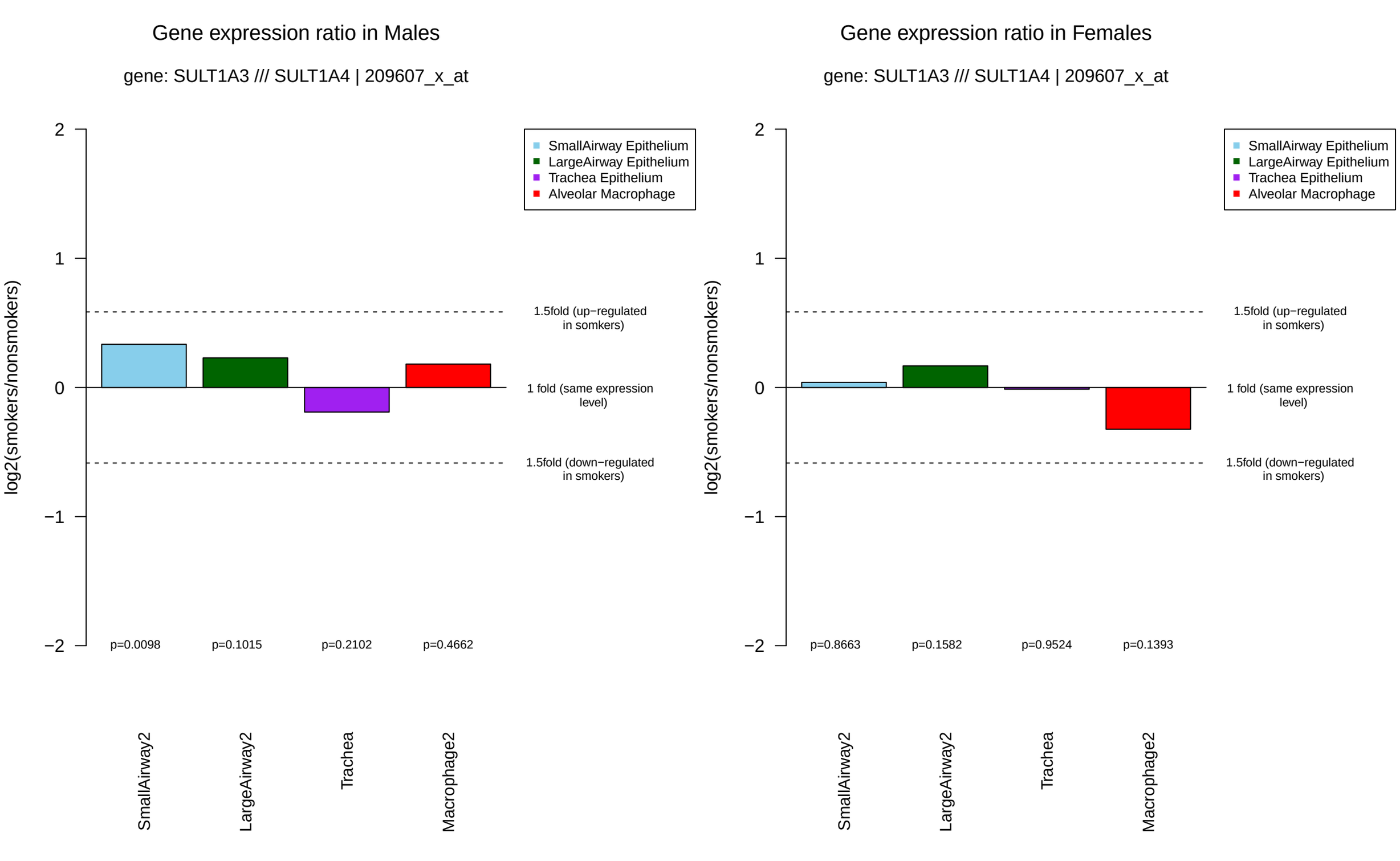
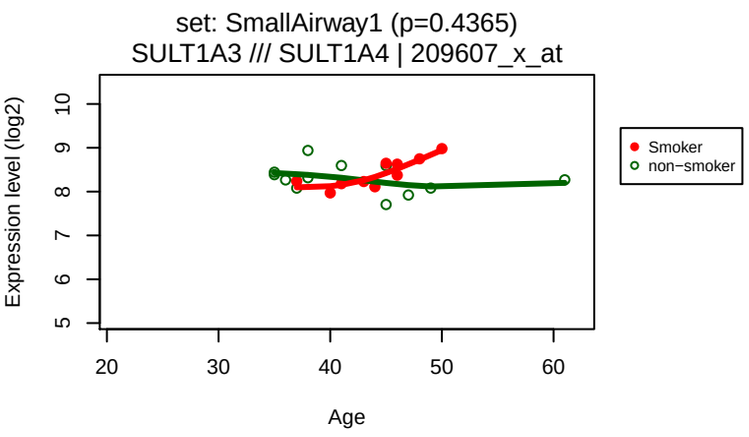
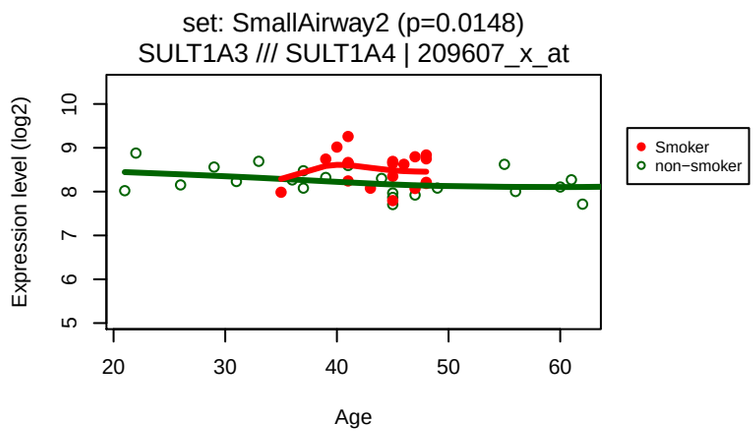
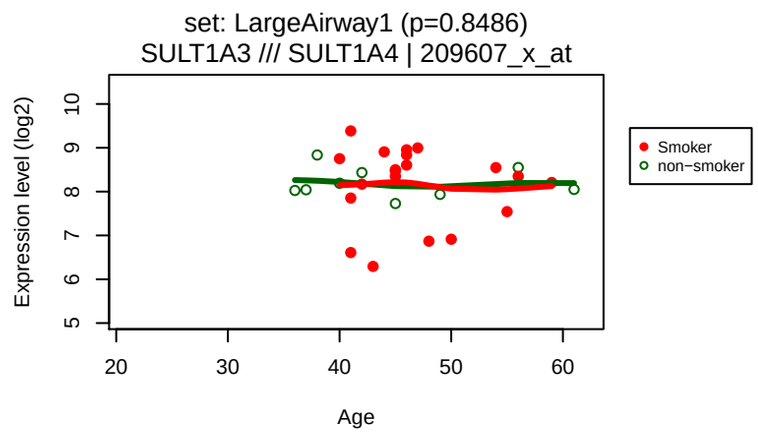

Gene expression of smokers and nonsmokers
gene: SULT1A3 /// SULT1A4 | 209607_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4365
p=0.0148
p=1e−04
p=0.8486
p=0.0688
p=0.2073
p=0.2535
p=0.8521
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

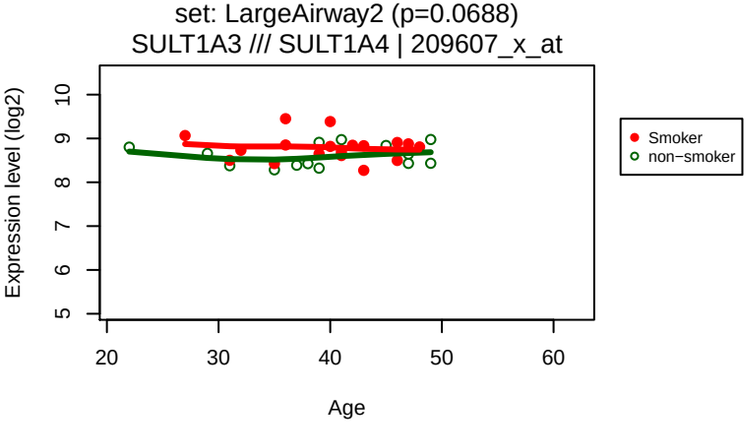

Gene expression ratio (smokers/nonsmokers)
gene: SULT1A3 /// SULT1A4 | 209607_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4365
p=0.0148
p=1e−04
p=0.8486
p=0.0688
p=0.2073
p=0.2535
p=0.8521
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

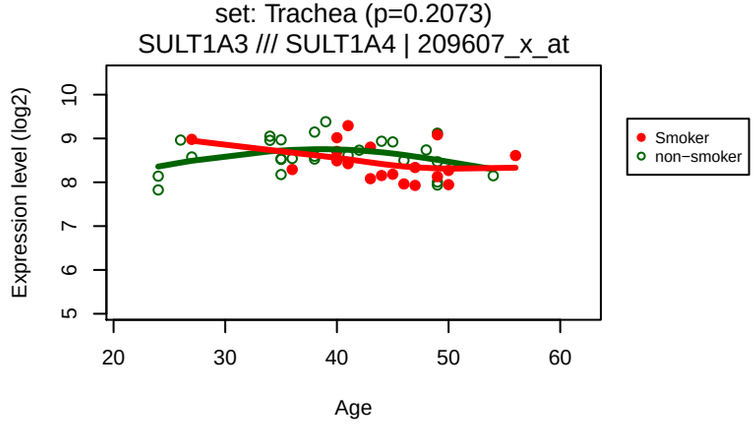

Gene expression ratio in Males
gene: SULT1A3 /// SULT1A4 | 209607_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0098
p=0.1015
p=0.2102
p=0.4662
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SULT1A3 /// SULT1A4 | 209607_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8663
p=0.1582
p=0.9524
p=0.1393
SmallAirway2
LargeAirway2
Trachea
Macrophage2

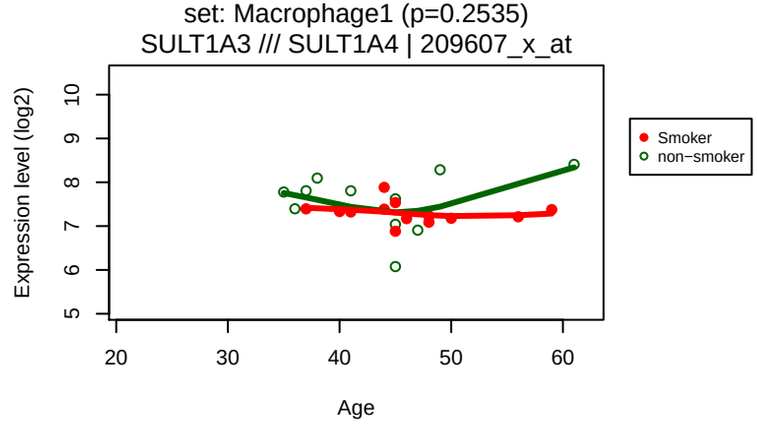

set: SmallAirway1 (p=0.4365)
SULT1A3 /// SULT1A4 | 209607_x_at
Expression level (log2)
Age
Smoker
non−smoker

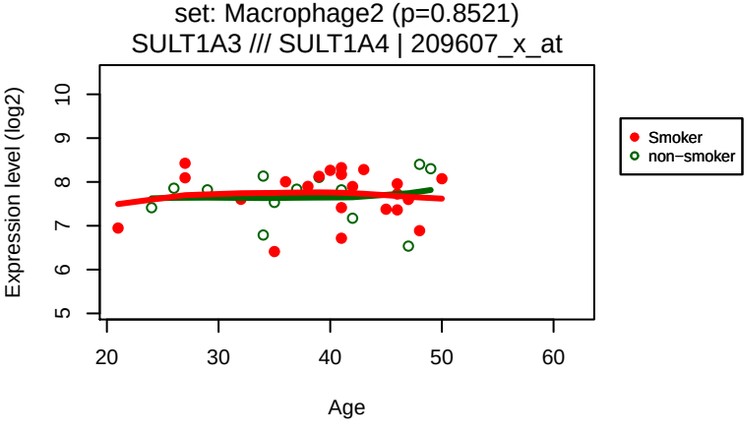

set: SmallAirway2 (p=0.0148)
SULT1A3 /// SULT1A4 | 209607_x_at
Expression level (log2)
Age
Smoker
non−smoker

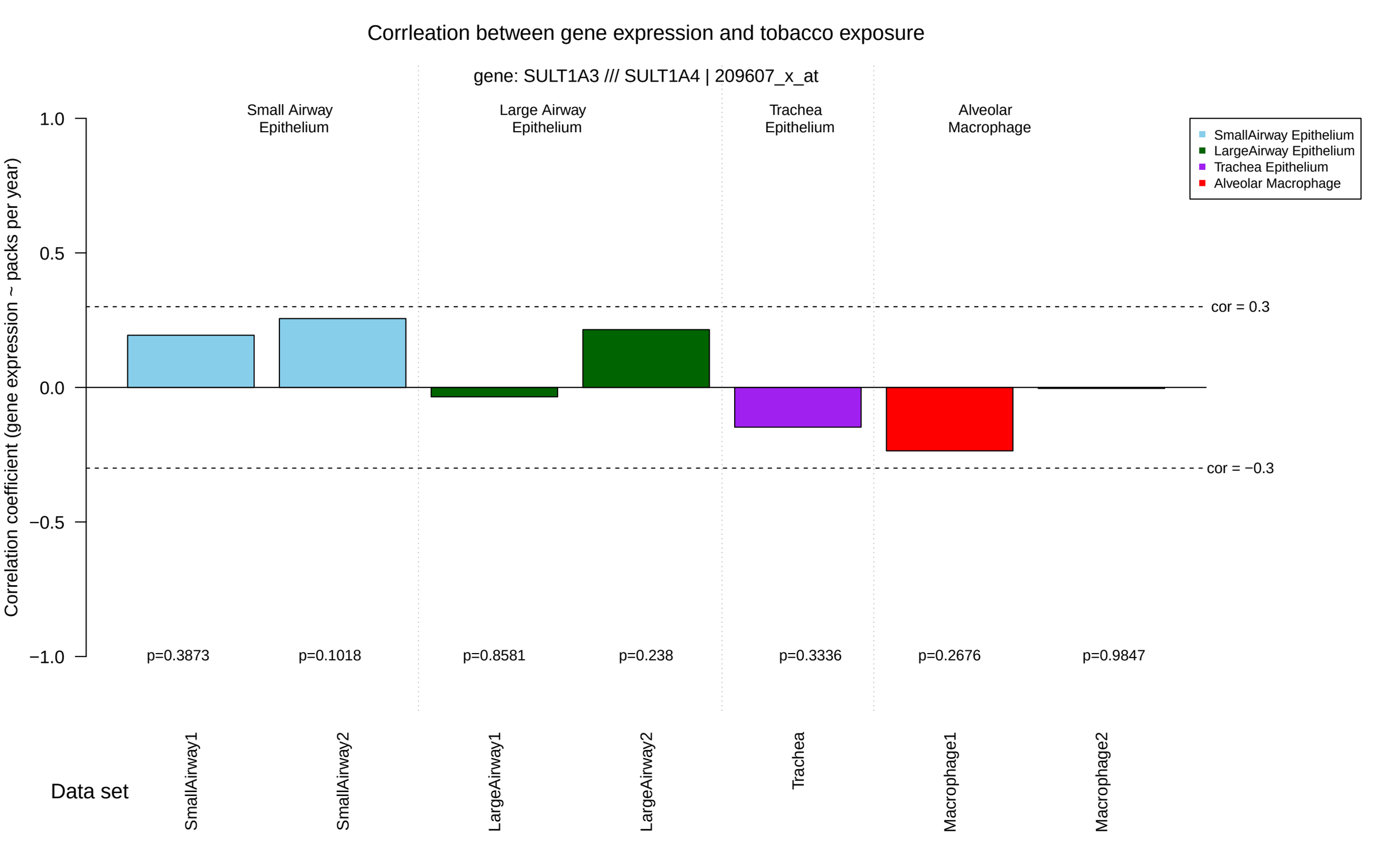

set: LargeAirway1 (p=0.8486)
SULT1A3 /// SULT1A4 | 209607_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0688)
SULT1A3 /// SULT1A4 | 209607_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2073)
SULT1A3 /// SULT1A4 | 209607_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2535)
SULT1A3 /// SULT1A4 | 209607_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8521)
SULT1A3 /// SULT1A4 | 209607_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SULT1A3 /// SULT1A4 | 209607_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3873
p=0.1018
p=0.8581
p=0.238
p=0.3336
p=0.2676
p=0.9847
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2